Jason Wang

wjas@umich.edu linkedin.com/in/wangjas github.com/jwang-um

Education

University of Michigan **May 2028**
BSE in Computer Science *Ann Arbor, MI*

- **GPA:** 4.0/4.0
- **Courses:** Data Structures & Algorithms, Combinatorics, Stochastic Processes, Matrix Methods for SIPML, Probability Theory, Machine Learning, Partial Differential Equations, Real Analysis, Linear Algebra

Experience

Rivian Automotive **May 2026 – August 2026**
ESS Data Science Intern *Irvine, CA*

- Prototyped agentic DTC-to-firmware mapping workflow, adopted as basis for a broader battery diagnostics initiative.
- Built reusable **Databricks Spark pipeline** unifying four telemetry formats, retained as team prognostics foundation.
- Delivered production **XGBoost** model for voltage sense wire faults, **300x PR-AUC lift** over prevalence baseline.
- **Reduced compute cost 4x** on core cleaning and query operations by restructuring Spark and pandas workloads.
- Evaluated OCSVM, Autoencoder, CNN-Transformer, and ROCKET models against handcrafted waveform features.

U-M Solar Car Team **July 2025 – Present**
Microsystems Engineer - Race Electrical Team *Ann Arbor, MI*

- Developed firmware for an **active aerodynamic fin**, optimizing control algorithms to maximize thrust.
- Validated **adaptive Kalman filters** and **Gaussian process regression** for anemometer denoising on telemetry.
- Built a **stochastic microscale wind simulator** for controls using forecast data and spectral turbulence models.
- Engineered **thread-safe CAN and UART APIs** in C, adopted as team's **standard hardware abstraction layer**.
- Refactored CAN-to-RF bridge firmware, implementing robust thread management using **FreeRTOS**.

Dare Auto Inc. **June 2024 – August 2024**
Software Validation Intern *Plymouth, MI*

- Standardized validation workflows by authoring functional test specifications for electro-hydraulic steering systems.
- Verified compliance of **10+ software iterations** via **HIL simulations** of diverse ECU network topologies.
- Engineered **Vector CANoe environments** with driver simulation algorithms to enhance system modeling.
- Programmed **Remaining Bus Simulations** in C and CAPL to simulate complex automotive ECU networks.

Projects

Music Analysis Pipeline | *Python, librosa, scikit-learn, PySide6*

- Designed **motif discovery pipeline** using STUMPY matrix profile and **DTW + agglomerative clustering**.
- Implemented structural segmentation via **spectral clustering** on chroma self-similarity matrices.
- Applied **NMF** for textural decomposition of audio files, automatically cross-validated against segmentation outputs.

Cogni: Knowledge Health Graph (MHacks 2026) | *TypeScript, Python, SpacetimeDB, Next.js*

- Deployed functional **Glean-style enterprise search** that flags outdated or conflicting docs and routes fixes to owners.
- Built **SpacetimeDB** knowledge graph supporting Drive, Confluence, Jira, Slack with embeddings, **5s sync latency**.
- Optimized **17 detectors** with indexed delta checks and **cached graph traversals**, flat cost at 5k docs.
- Developed Next.js/React UI with Tailwind, Auth.js, react-force-graph, deployed via **Docker Compose** on **AWS EC2.**

Cross-Scanner Brain Age Prediction | *Python, PyTorch*

- Trained **domain adversarial neural networks** to predict brain age from MRI, handling distribution shift across scanner types; course research project for EECS 545 (graduate Machine Learning).

Technical Skills

Languages: C, C++, Python, TypeScript, Java, MATLAB, Verilog, SQL, CAPL
Tools: Git, Linux, Databricks, ModelSim, STM32CubeIDE, Altium, Docker, AWS EC2
Technologies/Frameworks: PyTorch, scikit-learn, Spark, pandas, NumPy, React, Next.js, SQLite, FreeRTOS, MLflow
Concepts: Machine Learning, Time-Series Modeling, Anomaly Detection, Signal Processing, Knowledge Graphs, Distributed Data Processing, Agentic Engineering, Firmware Development